Change Request Form

I. General Information

Project name: Future Tech Network_
Project number: _243_
Requestor name: _Penny Nickles_
Requestor's contact information: PennyNickles@gmail.com_
Date of request: 12/20/2018_
Change request tracking number: 445845_
Date request approved/denied: 12/22/2018_

Section One—To Be Completed by the Requestor

II. Description of Change Request	*Include a detailed description of the requested scope change.*
The Future Tech Network needs to cover 100% of the first floor rather than just the north and south wings. Users should be able to access the network from anywhere in the building with no dead zones.	

III. Business Justification for Change	*Describe how the business, project, or product will benefit from the requested change.*
Allowing the network to reach all corners of the facility will not limit our employees to staying at their stations to get work done. Being that the first floor will consist primarily of sales employees and many of them will have laptops we want them to be able to walk and work. Providing our sales team with the tools they need is necessary to increase business and keep the company moving forward.	

IV. Impacts of Not Making the Change	*Describe how the business, project, or product will be impacted if the change is not made.*
If this change is not implemented the company runs the risk of losing sales. This could occur if an agent needs to go across the building to consult with other team members or managers during the bidding process. It is a high possibility that if the agents must end the call before the sale is complete and call the potential customer back the customer could be checking out competitors and go a different direction. While we want customers to be informed before finalizing sales we do not want to give them time to choose a competitor over Future Tech.	

V. Alternatives to Change.
Describe any known alternatives to the change.

Alternatives to this may be to limit the dead zones to the only the bathrooms and conference rooms. Ideally, paths should be configured to allow the sales team to travel with their laptop connected from one side of the building to the other without interruption.

Section Two—To Be Completed by the Project Manager

VI. Impacts of the Change — *Describe the impacts of this change to the project schedule, budget, and quality.*

Adjusting the network to cover the entire first floor would involve increasing the budget to account for the newly required access points, switches, and rj45 jacks. This would also delay the project's schedule by a minimum of one week as the newly added equipment would need to be ordered, delivered, installed, configured, and tested. The quality of the network would increase being that it could be accessed anywhere in the building; however, this increase would not be immediately compensated for by the little amount of sales that could potentially increase.

VII. Alternatives to Change — *Describe alternative solutions to the change.*

An alternate change to the network would involve arranging the equipment to allow the sales team members to cross the building without losing connection. This would be a single path that employees would be required to take from the north wing to the south to avoid interruption of service.

VIII. Recommendation to the Change Control Board
Describe the project manager's recommendation to approve or deny the change. Include justification for the recommendation.

It is my recommendation that we approve of the suggested change alternative. Accommodating the sales team so that they may cross over to either side of the facility without interruption will allow the organization to not limit sales and provide the least minimal changes to the current network design. This will also make it easier to transition if later on the organization wishes to provide full access to the network on the first floor or even reach outdoors.

Section Three—To Be Completed by the Change Control Board
Recommendation: *Include a discussion concerning the goals of the project. Does this change significantly impact the goals of the project such that the project goals can no longer be met?*
The goal of the Future Tech Network is to allow the organization to expand on its current reach. It is because of this that this change falls into expanding on reaching and obtaining a larger customer base.
Should this change be considered a new project or phase two of the existing project, etc.?
If obtainable without project delays, arranging the network to allow for a path of connectivity between the north and south wing of the facility can be added to the current project's agenda. However, due to the added expenses, increasing the network coverage to allow for 100% connectivity will be up for consideration as phase two of the Future Tech Network. Additional features may be added to phase two at a later date should new needs come about.
Date of Review:
12/21/2018
Signature:
Felicia Ainslie

Source: Helman, K. (2011). Project management jumpstart (3rd ed.). Indianapolis, IN: Wiley.

Proof of Update

Scope:

IV. Comprehensive List of Project Deliverables	*These are the products or services that must be produced in order to fulfill the goals of the project. Deliverables should have measurable, verifiable results and outcomes. Identify critical success factors. Will map roughly to the work package Level of the WBS*

This private internal network will span three floors and cover roughly 16,000 square feet of space. There will need to be roughly one access point for every 1,600 square feet for the top two levels and a minimum of two access points on the lower level (Kaleem, 2017). Routers and switches will be utilized to connect devices to the network and a modem will be used to allow access to the internet. Servers will also be utilized to store customer and organization sensitive data. Technical policies will be developed to ensure the network and organization's security. Additional security measures will be taken to ensure that only authorized individuals can access the devices on site, intranet, data, and servers. These will include but may not be limited to physical and virtual locks, firewalls, and anti-virus/malware software.

Change Request Form effective 12/20/2018:

The first level, or lower level, will have a minimum of three access points.

V. Comprehensive List of Project Requirements	*Requirements are the specifications of the deliverables.*

The finished network should provide each user with no less than 50mb download speeds and 5mb upload speeds. The top two floors will have absolutely no black out zones; however, the first floor only requires that there are no black out zones on the north and south wings of the building. All desktops, laptops, and servers will require physical locks in addition to virtual lock screens in place. All desktops and laptops will also require a notification upon being powered on to inform the user that they are subject to persecution under law should illegal or otherwise banned activities occur during use of the device. Prior to employment at the new location employees must sign a contract that informs them of what acceptable behavior is when using the devices on site as well as any other technical restrictions they must follow.

Change Request Form effective 12/20/2018:

The first level will include a path from the north and south wings to allow for continuous network connectivity. This will allow users to connect to the network and walk from one end of the building to the other without entering a black out zone.

VII. Time and Cost Estimates	*Include initial estimates of time and resources. These are estimates only and will be updated after additional project planning activities are completed.*

This project budget will include computer costs for roughly 500 employees who will not be sharing devices for security reasons. Assuming bulk discounts are applied, and a mixture of laptops and desktops will be purchased, the estimated cost for computers alone will be $250,000 (Thompson, 2017). Installing these devices along with the appropriate security measures will take roughly two weeks. Network security software will need to be installed on each of the company owned devices. As an estimate, Avast offers businesses a one-year contract for 200 devices at just over $5,100 a year which would be roughly $8,200 for all 500 devices (Avast Software, Inc., 2018). Three servers will also need to be installed and will take roughly one week to setup with a cost estimate of $1,800 (Gildred, 2018). Cabling will need to be installed throughout the building to allow the computers and servers to connect to the network. Installing the cabling for this network will take roughly three weeks and cost roughly $7,000 if Cat7 cabling is used (Fixr, 2018). Outside of the network cabling all other network hardware will take roughly one week to install. Additional costs will include the 22 access points for users to connect to the network. These access points will cost a little over $3,500 and possibly more after taxes and after the network speeds are tested (Linksys, 2018). This may also be reduced once routers are in place. An estimate of six routers will be needed for the top two floors and two for the first floor due to concrete barriers in the building which will cost roughly $2,800 (Hyde, 2018). Assuming that each user needs 2 ports to connect a computer and monitor a total of 1,000 ports will be needed for users to connect to the network. Taking this into account the price for the switches needed will cost a minimum of $67,000 (ALLHDD, 2018).

Change Request Form effective 12/20/2018:

Adjusting the first floors network configuration will add roughly two days onto the schedule. Taking this into consideration it will take approximately four and a half weeks to install and configure the network.

IX. Assumptions *List all project assumptions.*

- Installing computers, software, and security measures will take two weeks.
- Installing the servers will take only one week.
- Installing the cabling will take three weeks.
- Only 22 access points and 8 routers will be needed to give the employees the desired network speeds.
- All devices will be compatible with one another and the software installed.
- All employees will be available during their scheduled work hours.
- All vendors will fulfill their contractual obligations.
- The necessary resources to complete this project will be available when needed.
- Communication throughout the project will occur daily between team members.
- All stakeholders will attend the required meetings.
- Penetration has been completed to ensure that the appropriate security measures are in place.

Change Request Form effective 12/20/2018:
Adjusting the first floors network configuration will not require additional equipment but rather equipment to be rearranged.

X. Product Acceptance Criteria *List all project acceptance criteria.*

- The top two floors provide 100% network coverage.
- The first floor provides 100% network coverage to the north and south wings.
- The network speeds provide each user 50 Mbps upload and 10 Mbps download.
- There are separate servers dedicated to critical employee data and for customer data.
- Each computer is equipped with a warning message regarding device use when powered on.
- Each computer is equipped with an antivirus software.
- Each computer has the appropriate firewall rules and ports available.
- Users are able to begin testing their devices functionality by December 1st.
- Full network testing has been documented to ensure functionality to employees.

Change Request Form effective 12/20/2018:
The first floor will provide 100% network coverage to the north and south wings as well as a path of full connectivity to and from either wing.

Charter:

III. Project Objectives	*Describe the overall objectives of the project and what factors will determine the success of the project.*

The objectives for this project are to provide each user with a personal computer that can access the network and share data within the organization safely. Furthermore, employees will need to be able to work simultaneously at a speed that allows them to assist customers in real time. Determining factors in the success of this will include each user's ip address having a minimum speed of 50 Mpbs download and 10 Mpbs upload. Additionally, the organization will require that there is no more than 0.01% downtime per year (Kumar, 2015). Supplementary determining factors in this project's success will involve where the network can be accessed. The top two floors will need to have access to the network everywhere; however, the first floor needs only the north and south wings of the facility to be completely accessible. Regarding security success measures there will need to be antivirus software, password restrictions, and security policies put in place.

Change Request Form Effective 12/20/2018:
The first floor will now require network access to employees when crossing the facility. This will not cover the entirety of the east and west wings but rather a straight path to and from the north and south wings.

VI. Resource Costs and Estimates	*Provide cost estimates if known, including monies already expended such as a feasibility study or consulting time.*

The cost estimate for the hardware and software necessary to complete this project will be close to $340,000 (ALLHDD, 2018; Avast Software, Inc., 2018; Fixr, 2018; Gildred, 2018; Hyde, 2018; Linksys, 2018; Thompson, 2017). Assuming Future Tech Inc. is only allowing for hourly wages to be paid at the median average the estimate to be paid out in total will be roughly $37,200 (PayScale, Inc., 2018; Salary.com, 2018; Time, n.d.). This estimate also considered that the hours that will be put in by each team member involved during development. These hour assumptions included 400 hours by the project manager; 320 by the agenda director; 120 hours for the network engineer; and 80 hours for the systems analyst, software engineer, and testing engineer. These hours worked will include research, development, implementation, and testing time.

Change Request Form Effective 12/20/2018:
Due to the additional 12 hours that will be spent adjusting the network design on the first floor the network engineer's hours will be an estimated total of 132 hours. Accordingly, this puts the total salary estimate at $37, 632.

WBS:

G64

	A	B	C	D	E	F	G	H	I
1	WBS Number	Task Name	WBS Description	Must Start by Date	Must End by Date	Task Level of Effort	Task Duration (in days)	Predecessor (Finish-to-Start Dependency)	Planned Start Date
58	5	Adjust First Floor Design	Ensure first floor can walk between north and south wings uninterupted.	12/26/2018	12/31/2018	12	2	3.1, 3.2, 3.3, 3.4, 3.5, 3.6	12/24/2018
59	5.1	Access Points	Place access point to allow for continual connectivity through the hall between the north and south wings of the facility.	12/26/2018	12/31/2018	12	2	3.1, 3.2, 3.3, 3.4, 3.5, 3.6	12/24/2018
60	6	Test	Thorough testing needs to be completed to ensure devices, software, and network equipment are functioning appropriately.	12/20/2018	12/26/2018	80	5	1, 2, 3, 4	12/20/2018
61	6.1	Ensure network functionality	Devices need to be successfully connected to the network.	12/20/2018	12/23/2018	40	5	3.7	12/20/2018
62	6.1.1	Ensure first floor hall network functionality	Ensure that devices connect to network in the hall connecting the north and south wings.					5	
63	6.2	Test network speeds	Devices need to connect quickly to the network and run quickly using the network.	12/20/2018	12/27/2018	20	4	3.7	12/20/2018
64	6.2.1	Test first floor hall network speeds	Ensure that devices get adequate speeds in the hall connecting the north and south wings.					5	
65	6.3	Test network coverage	Devices need to be able to connect from anywhere on the top two floors and from the north or south wings on the first floor.	12/20/2018	12/31/2018	20	4	3.7	12/20/2018
66	6.3.1	Test North to South Hall Network Coverage	Test that network covers a straight path to/from the north and south wings of the first floor. (Change Request Form effective 12/20/2018).					5	
67	6.4	Test Security Measures	Ensure security measures prevent harm to devices and data.	12/20/018	12/31/2018	20	4	3.7	12/20/2018
68	7	Launch	The project needs to be finalized and the network fully available to all employees at the new Future Tech Inc. location.	1/1/2018	1/4/2018	30	2	1, 2, 3, 4	1/1/2018

Budget:

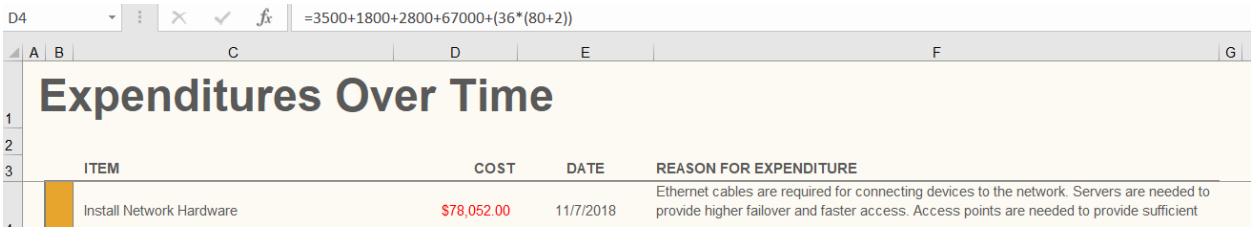
D49 =D8+D18+D20+D21+D22+D23+D24+D29+D30+D31+D32+D33+12

Source of Project Cost

	PROJECT TASKS	LABOR HOURS	LABOR COST ($)	MA
SALARIES COST	Project Manager	376.0	$40.00	
	Agenda Director	171.0	$30.00	
	Systems Administrator	58.0	$25.00	
	Network Engineer	260.0	$36.00	
	Software Engineer	73.0	$37.00	
	Testing Engineer	228.0	$29.00	
	Subtotal	**1,166.0**	**$197.00**	

Subtotals	**911.0**	**$39,845.00**
Risk (Contingency)	40.0	$3,040.00
Total (Scheduled)	**951.0**	**$42,885.00**

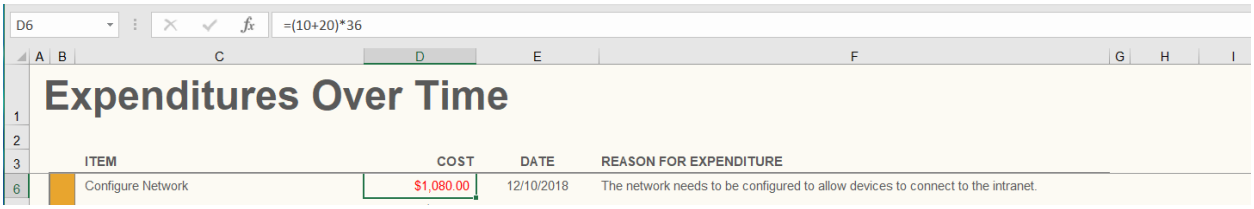
Source of Project Cost | Expenditures Over Time | Cumulative Project Costs | Data Worksheet



D4 =3500+1800+2800+67000+(36*(80+2))

Expenditures Over Time

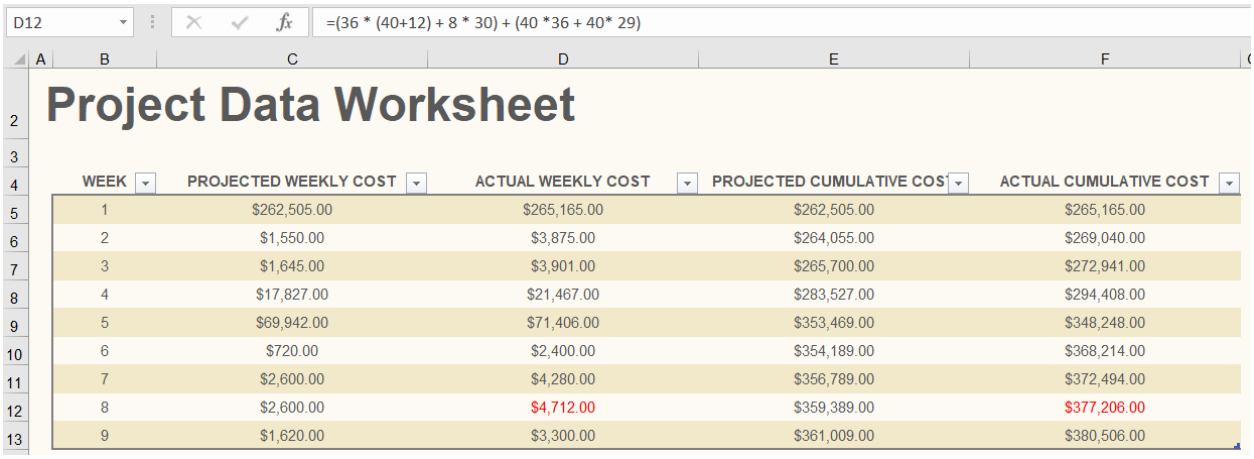
ITEM	COST	DATE	REASON FOR EXPENDITURE
Install Network Hardware	$78,052.00	11/7/2018	Ethernet cables are required for connecting devices to the network. Servers are needed to provide higher failover and faster access. Access points are needed to provide sufficient



D6 =(10+20)*36

Expenditures Over Time

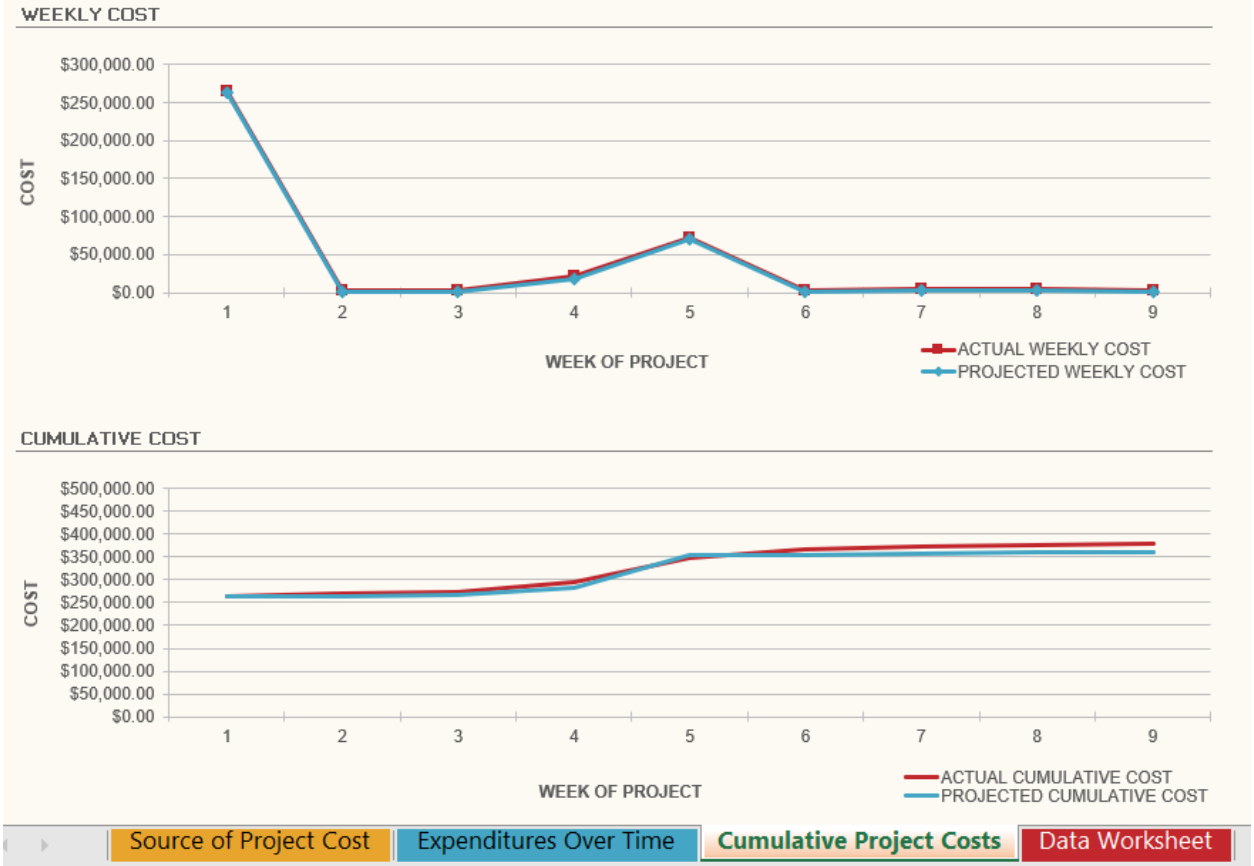
ITEM	COST	DATE	REASON FOR EXPENDITURE
Configure Network	$1,080.00	12/10/2018	The network needs to be configured to allow devices to connect to the intranet.



D12 =(36 * (40+12) + 8 * 30) + (40 *36 + 40* 29)

Project Data Worksheet

WEEK	PROJECTED WEEKLY COST	ACTUAL WEEKLY COST	PROJECTED CUMULATIVE COST	ACTUAL CUMULATIVE COST
1	$262,505.00	$265,165.00	$262,505.00	$265,165.00
2	$1,550.00	$3,875.00	$264,055.00	$269,040.00
3	$1,645.00	$3,901.00	$265,700.00	$272,941.00
4	$17,827.00	$21,467.00	$283,527.00	$294,408.00
5	$69,942.00	$71,406.00	$353,469.00	$348,248.00
6	$720.00	$2,400.00	$354,189.00	$368,214.00
7	$2,600.00	$4,280.00	$356,789.00	$372,494.00
8	$2,600.00	$4,712.00	$359,389.00	$377,206.00
9	$1,620.00	$3,300.00	$361,009.00	$380,506.00


WEEKLY COST
$300,000.00
$250,000.00
$200,000.00
$150,000.00
$100,000.00
$50,000.00
$0.00
COST
1
2
3
4
5
6
7
8
9
WEEK OF PROJECT
ACTUAL WEEKLY COST
PROJECTED WEEKLY COST
CUMULATIVE COST
$500,000.00
$450,000.00
$400,000.00
$350,000.00
$300,000.00
$250,000.00
$200,000.00
$150,000.00
$100,000.00
$50,000.00
$0.00
COST
1
2
3
4
5
6
7
8
9
WEEK OF PROJECT
ACTUAL CUMULATIVE COST
PROJECTED CUMULATIVE COST
Source of Project Cost
Expenditures Over Time
Cumulative Project Costs
Data Worksheet